

Research Proposal: Games-based Library Services Study & Needs Assessment

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INFOST 799-201: Introduction to Research Methods in Information Studies

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May 10, 2024

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Abstract

This research project proposes an in-depth investigation into the role and impact of games-based library services within public libraries throughout the United States. By employing two distinct questionnaires (one targeting library staff responsible for implementing games-based services – like library programs and collections – and another aimed at the library patrons said services are designed to engage), the study seeks to gain a comprehensive understanding of the both the current landscape of games-based services in American public libraries as well as respective libraries' community response to them. The project's ultimate goal is to generate valuable data-driven insights that might guide advocacy for and the development and enhancement of such services, ensuring that libraries might more effectively meet the modern needs and interests of a contemporary patron-base. Alternatively, if the study demonstrates that games-based library services are not especially desired by American library communities, the data will help libraries avoid unnecessary time and resource allotment toward implementing such services. In either case, outcomes of this research project could significantly assist librarians in making informed decisions regarding games-based library services.

Introduction

As technological innovation continues to occur at a seemingly exponential rate while widespread accessibility to (and interest of) modern media has become the norm for American households, many public libraries have found themselves in the position of needing to evolve, enhance, and diversify their library services in order to better meet the contemporary needs and interests of their patron base. While American libraries' collections and programs are rife with offerings and experiences related to books, movies, arts & crafts, and more, comprehensive data about multiformat games-based library services is remarkably absent, despite modern gaming's overwhelming popularity.

This lack of games-based knowledge for the field could have serious and lasting consequences when it comes to serving the modern library patron. A recent comprehensive report released by the American Library Association titled *Gen Z and Millennials: How They*

Use Public Libraries and Identify Through Media has indicated that as many as 53% of Gen Z and Millennial users identify as “gamers” (Inman Berens & Noorda, 2023). What’s more, the report indicates that these gaming library users are also avid readers, which presents an immense circulation and service usage possibility for librarians; this is epitomized by a quote of Dr. Noorda’s in the report, where she states:

“Great news: Younger generations of people are reading books, buying books, and visiting libraries,” ... “Not only are Gen Z and Millennials engaging with books, but they are also engaging with other forms of media. They are gamers, readers, writers, and fans who are comfortable with malleability between media categories and forms.”

The report outlines the importance of diversifying library media offerings further, outlining that: Libraries are drawing even Gen Z and Millennials who don’t self-identify as readers. More than half of the 43 percent of Gen Z and Millennials who don’t self-identify as readers have been to their local library in the past 12 months. “These digitally-immersed generations make clear that libraries are about more than books,” said ALA President Emily Drabinski. “Programming relevant to teens and their parents – coding clubs, job application help, gaming – draws even non-readers to the library, as does the physical space to connect and collaborate.”

Ultimately, the influence of more appropriately reaching Gen Z and Millennial library users will have far-lasting impacts beyond simply affecting current generations of patrons; these users will make families of their own, bring them to the library, thus creating the future ‘library users of tomorrow’ – library users who will have potentially grown up appreciating the diverse and cutting edge media options their libraries opted to offer, resulting in a positive feedback loop of library appreciation and investment.

This research project proposal aims to ameliorate the current games-based library services knowledge gap by effectuating an intricate process of relevant data acquisition and analysis. The project would include a detailed look at both the current landscape of games-based library service implementations in American public libraries as well as their reception within participating libraries’ communities. Study outcomes would keenly illuminate the status of modern gaming as offered in public libraries, gauge patrons’ interest and desire for such services,

and assess games-based library service usage, thereby providing valuable insights for future libraries' service development, budgeting, policy-making, and more.

Functionally, the research would be structured around two dependent variables: "community interest in games-based library services" and "usage of games-based library services". These variables are examined in relation to two independent variables: "libraries' current games-based library services offerings", and "community/patron demographics". By investigating the relationships between these variables, the study seeks to answer pivotal questions about both the frequency and rigor of games-based library services as they relate to general patron desire for these types of services, and any effect that demographic identifiers might have in relation to usage and interest.

Population-wise, the study will investigate these variables within the scope of two key groups: public library staff involved in service planning and management, and public library patrons who engage with or could potentially engage with games-based library services. The research design (outlined more thoroughly later in this paper) targets a substantial sample size of library staff and users throughout the United States; the ambitious scope of the study's subject population goal would ensure diverse representation of staff and patrons from different regions, library sizes, and more. Robust data collection and analysis are central to the study's research; the study's wide-reaching data will be subject to comprehensive analysis that will employ both descriptive and inferential statistics examinations.

In the end, the study's findings promise to result in informative conclusions that would be directly useable by practicing library staff to better serve their patrons, significantly impacting public libraries' decisions regarding games-based library services – all while meaningfully contributing to the broader field of library and information science research as a whole. Directly put: public libraries need to examine more thoroughly (and in all likelihood, integrate) games-based library services; this study could serve as a foundational project to help public libraries meet this goal.

Research Problem

The study's research problem centers around the lack of extensive research into implementation metrics and reception of games-based library services in public libraries throughout the United States. Despite the all-time high (and ever-growing) popularity of modern

games-based mediums (like video games, board games, tabletop role-playing games, and collectible/competitive card games), there is a significant gap in our understanding of how thoroughly games-based services are being utilized in American public libraries, and if/how games-based services are desired within their communities. Additionally, professional library education/training programs lack core learning opportunities surrounding games-based library services integration (Elkins & Hollister, 2020).

This lack of knowledge poses a barrier to the development and improvement of games-based library services that might effectively cater to contemporary community needs and interests; if libraries do not properly research and meet modern media/service interests of library patrons, they run the risk of experiencing declining usage, and being seen as ‘irrelevant’ in the eyes of their patrons (and funding bodies) (Adams, 2009). On the other side of the coin, if libraries impulsively opt to spend time and money developing robust games-based library programs and collections without investigating patrons’ actual desire for them, they run the risk of wasting precious resources on unpopular offerings. This proposed study would result in valuable information that could help amend both of these potential concerns.

“Games-based Library Services” Defined

It’s important for those involved in the study to understand what exactly “games-based library services” [hereto referred to GBLS] encompasses. In the context of this study, “games-based” means any library experiences/offerings directly centered around modern gaming media (including formats like video games, board games, tabletop role-playing games [TTRPGs], virtual and augmented reality games [VR and ARGs], and collectible/competitive card games). “Services” refers to any library offerings intended to be utilized/experienced by library patrons (like in-person or virtual programs, circulating collections, drop-in/“library use only” collections, informational displays, take-home “passive program” kits, giveaways, and more).

As such, GBLS could be defined as “library experiences/offerings directly centered around modern gaming media like video games, board games, TTRPGs, VR and ARGs, collectible/competitive card games, and other related contemporary gaming formats.” In order to ensure that surveyees are informed about the concept’s meaning, GBLS will be properly defined in the respective questionnaires’ introductions. Furthermore, surveyees will be given contact

information to reach out to in order to seek clarification about study terminology in case there is any confusion.

Research Questions & Hypothesis

The study's research questions are as follows: 'What is the current state of GBLS offerings in public libraries?', and 'Is there a demand for more of said services in public library communities?'. Furthermore, 'Does the presence and variety of games-based library services correlate with some level of community interest and the frequency of service usage?'. Finally, 'How do patron demographics relate to interest in GBLS, if at all?'.

Related to these research questions, my hypotheses are:

1. There are fewer libraries that *prioritize* offering GBLS than those that do not; i.e., many libraries likely do not offer GBLS in any form, and most libraries that do offer them likely do not resource them with time and funding the way that they might be resourcing *non*-games-based library programs and collections – despite modern gaming's popularity.
2. Libraries that offer GBLS (and those that prioritize offering robust GBLS in particular) will report them as being very popular with their patron-base.
3. Patrons will report meaningful usage and an overall positive response to GBLS where libraries offer them; where libraries do not offer GBLS (or do not offer especially robust GBLS), patrons will report that they desire more of them.
4. Youth patrons, Teen patrons and Adult Patrons aged 18 to ≈ 40 will express significant interest in GBLS, while Adult Patrons aged $\approx 41+$ will not express as much interest.

Purpose of Study

The purpose of this study is to use both a library staff-facing "Games-based Services Implementation Questionnaire" (Appendix A) and a patron-facing "Games-based Library Services Needs Assessment Survey" (Appendix B) to gather response data that would provide a detailed understanding of the current state of GBLS in public libraries throughout America, their reception within surveyed libraries' communities, whether surveyed libraries' communities

desire more GBLS, and how library user demographics might correlate with GBLS interest/desire.

By investigating any correlation between the presence and variety of these services and the level of community interest/usage of them, the study aims to provide librarians with actionable data related to current implementation prevalence and patron desire of GBLS, which can help librarians advocate for funds related to developing and maintaining GBLS, while also aiding in GBLS-related programming/collection development decision-making processes. Correlations might indicate that certain types of GBLS are desired more than others, and/or that certain GBLS are more successful in terms of usage than others.

Alternatively, if the study demonstrates a lack of desire/need for GBLS as expressed by patrons, or an obvious lack of popularity by library staff already implementing them, libraries can use that information to allot their time and resources toward other types of programs and collections. In either case, study results will serve as a modern “snapshot” of the current status of games-based library services in America, and will be able to be used as a foundational project to be built upon with any future research in the space.

Literature Review

While robust GBLS surveying has not yet been done with the scale and precision this project aims to, there are still numerous journal articles, reports, blog posts, websites, recent studies, and more that add helpful context to the general exploration and efficacy of public library-centered gaming services. These pieces fall into multiple different library and/or games-related categories, including research about the beneficial learning principles of games, the popularity/pervasiveness of gaming, recent research/reports about modern library users and their interests/needs, successful researcher/library practitioner partnerships, best practices for games-based services integration (via programs and collections), and more. This paper will offer a brief outline and overview of some of the most useful, well-developed, contemporary, and accessible writings among the aforementioned categories.

Understanding the interests and needs of modern library users is crucial for the development of effective library services. Recent research from the American Library Association’s report titled *2022 Public Library Services for Strong Communities: Results from The 2022 PLA Survey* suggests that today's library users value a wide range of services, from

traditional book lending to digital resources and community programs (Goek, et al., 2023). They appreciate libraries as spaces for learning, creativity, and community engagement. Furthermore, as illustrated in Noorda & Inman Berens' 2021 article *Immersive Media and Books 2020: New Insights About Book Pirates, Libraries and Discovery, Millennials, And Cross-Media Engagement: Before and During Covid*, there is a growing interest in interactive and experiential learning opportunities, which games-based library services can provide. The needs and interests of library users can vary widely depending on factors such as age, education level, and community context; therefore, it is important for libraries to regularly assess the desires and interests of their users to ensure that their services remain relevant and effective. This crucial library services evolutionary concept is effectively highlighted in Dr. Abigail Phillips' and Victor Lee's 2019 book *Reconceptualizing Libraries: Perspectives from The Information and Learning Sciences*.

Considering these facts about modern library users, it would behoove librarians to understand just how utterly popular modern gaming (via multiple different formats) has become. The pervasiveness and enthusiasm surrounding gaming in modern society is undeniable; with the advent of digital technology especially, games have become a ubiquitous part of our culture (Howarth, 2024). Video games, board games, and collectible/competitive card games are played by people of all ages, from children to adults, and across various platforms, from analog options, to consoles, to computers, and to mobile devices; the video gaming industry in particular has seen significant growth among all formats, with revenues surpassing those of the film and music industries, as highlighted by the Entertainment Software Association's *2023 Essential Facts About the U.S. Video Game Industry*. It's not just about digital games, though, with games-based popularity extending to various genres and types games; recent articles by Jon Cherry (titled *What's Behind The Rise in Popularity Of Board Games?*, from 2023), Jaclyn Peiser (*Why Board Games Are Quickly Regaining Popularity*, 2024), and PlayToday.co's writer 'Florence' (*What Percentage of People Play Board Games*, 2024) all illustrate multi-modal games' popularity. The widespread pervasiveness of games suggests a high potential for their use in library services for years to come, as expressed by the American Library Association Games and Gaming Round Table on their webpage titled *Why Have Games in Libraries?* (ALA GameRT, 2024c).

In order for libraries to be successful, the integration of games-based services requires careful planning and implementation. Like Sandra Abrams emphasized in her 2021 article

Checked Out? Videogame Programming in U.S. Libraries and Implications for Videogame Literacies, best practices suggest starting with a clear understanding of the library's goals and the needs and interests of its users; the selection of games should align with these goals and interests. Libraries should also consider the logistics of implementing games-based services, such as the availability of space and resources, and any training for library staff that may be necessary to ensure that they are comfortable facilitating games-based activities. Carlie Forsythe's 2021 article *Game On! Recreational Play in the Library: Reflections of a Board Game Librarian*, Post et. al's 2023 article *A Colossal Catalog Adventure: Representing Indie Video Games and Game Creators in Library Catalogs*, and Robson, Phillips, and Guerro's 2018 article *Don't Just Roll the Dice: Simple Solutions for Circulating Tabletop Game Collections Effectively In Your Library* all do a stellar job of exploring such logistical needs – and their solutions. In an ongoing basis, libraries should regularly evaluate the effectiveness of their games-based services and adjust as necessary; Goodridge and Rohweder's 2022 book *Librarian's Guide to Games and Gamers: From Collection Development to Advisory Services* does an excellent job of compartmentalizing the work required to successfully integrate GBLS smoothly into library operations while continuously evaluating their success, allowing for effective long-term programming and collection revision. It's not always easy, but despite such challenges, Christa Hill outlines in her 2016 article *Play On: The Use of Games in Libraries* that the successful integration of games-based services can greatly enhance the library experience for users.

In advocating for stronger GBLS, it can be helpful for libraries to have language and research that backs up the social and emotional benefits that games can provide to their users (Farber, 2021). Fortunately, games as a medium have long been recognized for their potential to facilitate effective learning; one of the most foundational works on the subject is James Paul Gee's 2008 book *What Video Games Have to Teach Us About Learning and Literacy*. The interactive nature of games allows players to engage with content in a dynamic way, promoting active learning, critical thinking, situated experience, and more; games can provide immediate feedback, allowing players to adjust their strategies and learn from their mistakes. They can also foster problem-solving skills, as players must often navigate complex systems and make strategic decisions to achieve their objectives. Furthermore, games can promote collaboration and communication, as many games require players to work together to achieve common goals. All of these beneficial components are robustly explored by the author of this paper's previous work,

Video Games: Their Effect on Society and How We Must Modernize Our Pedagogy for Students Of The Digital Age (Baker, 2014). The motivational aspects of games, such as the desire to achieve high scores or complete challenges, can also encourage players to engage deeply with the content. These learning principles align well with contemporary pedagogical theories, which emphasize the importance of active learning, feedback, problem-solving, collaboration, and motivation in promoting effective learning. Some of the best literature on the intersection of learning, service, and libraries have been written by Jared A. Seay, the Reference & Instruction Librarian/Media Services Coordinator at the Addlestone Library of the College of Charleston in South Carolina; valuable articles of his include:

- *Considering Games and Gamification in Libraries & Associated Entities — In Praise of Real People* (2018)
- *Considering Games and Gamification in Libraries & Associated Entities — Surely, You Can't Be Serious: When Library Folk Go Game* (2018)
- *Considering Games and Gamification in Libraries & Associated Entities — The “G” Word* (2018)
- *Considering Games and Gamification in Libraries & Associated Entities — How the Longing for Tabletops Has Revitalized Games* (2018)
- *Considering Games in Libraries and Such — Space Guardians: New Year's Musings of How Libraries of Old (Including but Not Necessarily Old Librarians) Have Returned Just in Time to Save Us From Digital Anxiety* (2021)
- *Considering Games in Libraries and Such — The Ultimate Library Instruction* (2021)
- *Considering Games in Libraries and Such — A Liminal Response to Watching Paint Dry* (2021)
- *Considering Games in Libraries and Such — On Using a Game to Actually Teach Something* (2021)
- *Gaming & Libraries* (2024)

Finally, it's important to consider how successful partnerships between researchers and library practitioners could greatly enhance the quality and impact of library services. Such partnerships can facilitate the integration of research findings into practice, ensuring that library services are evidence-based. They can also promote the generation of new knowledge through

collaborative research projects. Furthermore, these partnerships can enhance the professional development of library practitioners, equipping them with the skills and knowledge to implement effective services. A second-to-none example of librarian/researcher collaboration can be seen in what is called the *PolarVR Project* (<https://polarvr.wisc.edu/>), which sees the partnership of “rural librarians, bilingual science communicators, polar scientists and a technical team to create a series of five bilingual virtual reality (VR) experiences, engaging users as polar explorers of exotic and extreme environments to enhance STEM understanding and appreciation.” Ponto, et al. explore the project thoroughly in their recent 2023 paper *Waddle: Using Virtual Penguin Embodiment as A Vehicle For Empathy And Informal Learning*.

Study Components, Research Procedures, and Analysis Strategy

For this study, there are two dependent variables. The first dependent variable is **Community Interest in GBLS**; this will be measured via relevant responses to specific survey questions about library users’ interest in different types of GBLS, and will help answer the question of whether American public libraries’ communities would like to see more GBLS implemented. The second dependent variable is **Usage of GBLS**; this will be quantified through questionnaire data on the frequency and duration of GBLS usage by community member respondents, and will help clarify whether GBLS are popular or not where they’re already implemented.

We will be considering two independent variables for the study, as well. The first independent variable is **GBLS Offerings**; e.g., the presence and extent of GBLS being offered in participating public libraries. This IV will help determine whether there is a correlation or not between libraries that frequently offer GBLS (or do not offer GBLS, or do not offer especially robust GBLS) and measured interest in them by their communities. The second will be **Community Demographics**; e.g., whether factors such as age, education level, and frequency of library usage (as reported in the patron questionnaire) affect desire for and usage of GBLS.

There are two populations for the study: **public library staff** involved in the planning, implementation, and management of library services (like collection development and program design/facilitation), and **public library patrons** – community individuals who engage with or have the potential to engage with GBLS. In order to ensure robustness of data and comprehensive library characterization within the results, this study would aim to utilize a

substantial sample size to analyze, with a target goal of surveying at least one (but preferably more) staff member(s) from approximately 1,700 public libraries across the United States. This ambitious number was determined based on the total number of public libraries in the country, which was identified as $\approx 17,278$ as of 2019 (ALA, 2019); thus, our target number amounts to approximately 10% of American public libraries, which would ensure a diverse representation from different regions and library sizes. For the patron survey, the study would aim to gather feedback from an average of 20 patrons from each participating library, resulting in an estimated total of 34,000 patron responses. The study's minimum survey goal would be 850 public libraries throughout the United States (5% of countrywide public libraries), and 17,000 patrons.

To reach such an aggressive number of libraries and patrons, the study would employ a leadership-based, "top down" promotion/communication approach by partnering with the Chief Officers of State Library Agencies (COSLA, 2024), and the American Library Association Games & Gaming Round Table via their various social media and networking channels (ALA GameRT, 2024a, 2024b) to nationally disseminate (and encourage) the call for participation. This tactic should result in sufficient library involvement and would provide the study with persuasive data.

Once the call for participation has resulted in an adequate number of interested libraries (who will have provided contact information via link to a basic form), the actual subject selection and data collection process would be completed; first, a stratified random sampling of varied public libraries across different states would be selected (to account for subgroup attributes like regional demographics, differently-resourced libraries, urban vs. metropolitan libraries, etc.), and program and collection development-responsible staff at these libraries would be asked to complete the Games-based Services Implementation Questionnaire. Those librarians would in turn be given simple instructions (via email) to offer the Games-based Community Needs Assessment Survey to a stratified random sampling of individuals within their communities (to account for subgroup attributes like age, education level, frequency of library usage, etc.).

Instrumentation for the study will include two Qualtrics questionnaires (also referred to as "surveys" to use common language for patrons) to collect data from library patrons and the staff that serve them. These questionnaires will be designed to gather detailed responses on the variety and depth of offered GBLS (from staff), and the level of community interest/usage of

these service (from patrons), respectively. Both questionnaires will collect demographic information for analysis.

The data acquired from the questionnaires will be organized using Microsoft Excel for readability (Microsoft, n.d.). Statistical software, such as IBM® SPSS® Statistics will then be used to thoroughly analyze the data in order to identify any trends and correlations between the presence and variety of GBLS and their level of community interest/usage (IBM, n.d.); the data collected from the questionnaires will be subjected to rigorous statistical analysis, and descriptive statistics will be used to summarize the data and provide a clear overview of the current state of GBLS offerings/usage in public libraries. Additionally, inferential statistics – including chi-square tests for categorical variables (Biswal, 2023) and t-tests for continuous variables – will be used to identify significant relationships and differences between study metrics. Regression analysis might also be employed to predict the long-term impact of certain factors on the interest and usage of GBLS (Simplilearn, 2024). Finally, the analysis will generate illustrative graphs and flowcharts to visually represent the findings, aiding in the interpretation and application of the study’s results/findings for libraries and library users nationwide.

Potential limitations that will could affect the study might include **response bias**, where respondents may not accurately report their interest or usage of services, and **limited generalizability** due to the scope of our sample size (being country-wide). Additionally, while a stratified random sampling approach aims to provide a representative sample, library staff and patron respondents may not accurately represent all mitigating factors and potential library communities; thus, the findings may not be applicable to *all* possible libraries considering the results. It should be noted that ensuring the ethical integrity of the study’s research is of paramount importance. All participants will be informed about the purpose of the study in the survey introduction and accompanying email/website descriptions. Participation is entirely voluntary, and all responses will be anonymous; while indirect identifiers will be asked for, no personally identifiable information will be collected. Additionally, data will be stored securely and only accessible to the research team. Any reports or publications resulting from the research will present aggregated data, ensuring that individual responses cannot be identified.

Ultimately, this study and process will offer tangible measurement metrics in the way of questionnaire majority responses, correlative response identifiers, and more, which will help us determine what affect our independent variables might purportedly have on our dependent

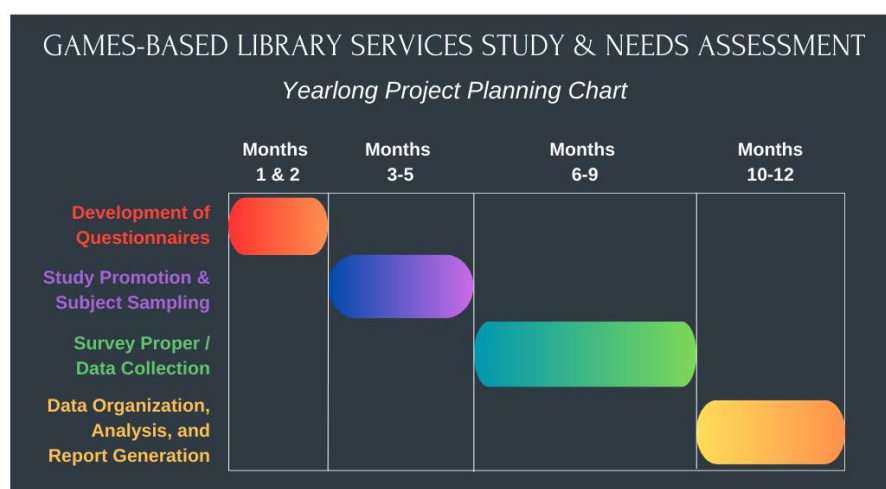
variables (if any), and whether our proposed hypotheses are rejected or not -- especially when considered in the scope of our research questions.

Project Schedule

It is anticipated that a year will be required to successfully complete the study. If more time is required than anticipated, the Principal Investigator will notify the funding body to account for unexpected time/costs.

- The first two months will be dedicated to the development and revision of the questionnaires [draft example surveys are included in **appendices A** and **B** at the end of this paper]. This will involve designing the questionnaires and testing them for validity and reliability with a small focus group.
- Months three through five will involve promoting the study and survey opportunity to acquire a viable library response pool, before stratified random sampling to identify exact library participants, and then finally distributing the questionnaires to the selected libraries.
- The following four months will be allocated for appropriate library staff and patron data collection, allowing ample time for respondents to complete the questionnaires.
- The final three months will be used for data organization/examination and report writing/distribution, during which the collected data will be analyzed, findings will be interpreted, and a comprehensive report will be written and disseminated.

The anticipated twelve-month schedule is illustrated in the following Gantt chart:



Itemized Budget (Draft)

The budget for this research project is relatively constrained and straightforward; the following numbers are estimated amounts that would likely cover each cost area for successful completion of the study as outlined above. Note that this budget accounts for staffing costs as being already relevant work to be integrated into Principal Investigator's existent job.

- **Questionnaire Development: \$3,000** – This includes costs for designing, testing, and refining the questionnaires.
 - **Data Collection: \$5,000** – This covers the distribution of questionnaires and the collection of responses.
 - **Data Analysis: \$4,000** – This includes costs for data organization and statistical analysis software.
 - **Report Writing: \$2,000** – This covers the time and resources needed to interpret the findings and write a comprehensive report.
 - **Travel and Lodging: \$4,000** – This budget line will help cover costs for meetings and presentations related to the project.
 - **Stipends for Participating Libraries: \$5,000** – While we will not be able to provide robust compensation to surveyee locations, this is to incentivize libraries to participate in the study.
 - **Publication/Documentation/Dissemination: \$2,000** - This covers the costs of making the research findings publicly available.
 - **Consultant Services: \$1,500** – This is for any unexpected external expertise needed during the project.
 - **Computer Services: \$500** – This covers any additional computer software costs or services needed.
 - **Administrative Costs for Partners: \$1,000** – This is to cover any overhead administrative costs incurred by partner organizations involved in the project.
- **TOTAL ESTIMATED COST: \$30,000**

Study Significance, Anticipated Implications, and Conclusion

The significance of a comprehensive community needs assessment and corresponding study of games-based library services integration cannot be overstated. In an era where information is increasingly digital and interactive, public libraries must evolve to remain relevant, responsive, and forward-thinking. They must adapt to the changing landscape of information dissemination and consumption, and GBLS offer a promising avenue for such adaptation.

The potential implications of this study are manifold:

- Firstly, the findings could significantly impact public libraries' programming, collection development, and other service-based decisions. By understanding the community's interest and usage patterns of GBLS, libraries can tailor their offerings to better meet the needs and preferences of their patrons. This could lead to heightened patron satisfaction and increased library usage, thereby reinforcing the role of libraries as vital community hubs.
- Secondly, the study could provide valuable insights into the current allocation of resources libraries use to fund and structure their programs and collections; if the patron surveys indicate meaningful interest in GBLS, libraries can use the information to advocate for stronger collection development and programming budgets related to gaming. Alternatively, if patrons express a lack of interest for GBLS, the findings could help librarians avoid spending precious time and resources on undesired program and collection development options. In either case, the study would allow libraries to focus their efforts on the services most appreciated and likely to be utilized by their communities, optimizing service delivery.
- Thirdly, the research would contribute significantly to the broader field of library and information science. Despite the growing popularity of gaming worldwide, there is a noticeable gap in library science-based literature (and, perhaps more important, professional library training/education) on the topic. This study could help fill this gap by providing empirical data on the interest and usage of GBLS in public libraries, stimulating further research in the area, and leading to a richer understanding of the role and potential of GBLS in library operations.

- Lastly, the study could have future-facing implications beyond the immediate context of public libraries; the findings could inform policy decisions related to library funding and support, and could open up new collaboration and partnership pathways with other institutions, organizations, and agencies within libraries' communities – particularly those looking to engage their audiences in innovative ways.

In conclusion, this study represents a significant and much-needed step towards understanding the role of GBLS in public libraries as well as their potential to enhance library offerings. By shedding light on community interest and usage of GBLS, the study would provide actionable and accessible data that to help libraries remain at the forefront of information services while meeting the evolving needs of their communities. As we navigate the complexities of the information age, studies like this one are crucial in guiding our path forward, ensuring that our public libraries continue to serve as dynamic, responsive, and inclusive community hubs.

References

- Abrams, S. S., & Van Eck, R. N. (2021). Checked out? videogame programming in U.S. libraries and implications for videogame literacies. *Journal of Adolescent & Adult Literacy*, 65(3), 227–236. <https://doi.org/10.1002/jaal.1197>
- Adams, S. S. (2009). The case for video games in libraries. *Library Review*, 58(3), 196–202. <https://doi.org/10.1108/00242530910942045>
- ALA GameRT. (2024a). *ALA GameRT Twitch Channel*. ALA GameRT - Twitch. <https://www.twitch.tv/alagamert>
- ALA GameRT. (2024b). *League of Librarian Gamers - ALA GameRT*. Facebook. <https://www.facebook.com/groups/LeagueOfLibrarianGamers/>
- ALA GameRT. (2024c). *Why have games in libraries?*. ALA Games and Gaming Round Table. <https://games.ala.org/why-have-games-in-libraries/>
- American Library Association [ALA]. (2019). *Libguides: Library statistics and figures: Number of libraries in the United States*. Number of Libraries in the United States - Library Statistics and Figures - LibGuides at American Library Association. <https://libguides.ala.org/c.php?g=751692&p=9132142>
- Baker, C. J. (2014, December 12). *Video games: Their effect on society and how we must modernize our pedagogy for students of the digital age* (thesis). *VCU Scholars Compass: Theses and Dissertations*. Virginia Commonwealth University. Retrieved from <https://scholarscompass.vcu.edu/etd/3627/>.
- Biswal, A. (2023, October 11). *What is a chi-square test? formula, Examples & Uses: Simplilearn*. Simplilearn.com. <https://www.simplilearn.com/tutorials/statistics-tutorial/chi-square-test>
- Cherry, J. (2023, April 26). *What's behind the rise in popularity of board games?* Cherryflava. <https://cherryflava.com/whats-behind-the-rise-in-popularity-of-board-games/>
- Chief Officers of State Library Agencies [COSLA]. (2024). *Cosla Overview*. Chief Officers of State Library Agencies. <https://www.cosla.org/overview>
- Elkins, A. J., & Hollister, J. M. (2020). Power up: Games and gaming in library and information science curricula in the United States. *Journal of Education for Library and Information Science*, 61(2), 229–252. <https://doi.org/10.3138/jelis.2019-0038.r1>

- Entertainment Software Association [ESA]. (2024). *2023 essential facts about the U.S. video game industry*. The Entertainment Software Association. <https://www.theesa.com/2023-essential-facts/>
- Farber, M. (2021). *Gaming sel: Games as transformational to social and emotional learning*. Peter Lang.
- Florence. (2024, March 6). *What percentage of People Play Board games?[board game statistics 2024]*. PlayToday.co. <https://playtoday.co/blog/stats/board-games-statistics/>
- Forsythe, C. (2021). Game on! recreational play in the library. *Emerging Library & Information Perspectives*, 4(1), 143–156. <https://doi.org/10.5206/elip.v4i1.13442>
- Gee, J. P. (2008). *What video games have to teach us about learning and literacy*. Palgrave Macmillan.
- Goek, S., Jones, K., Clark, L., Drill, K., & Price, J. (2023). *2022 Public Library Services for strong communities: Results from the 2022 PLA Survey*. American Library Association. https://www.ala.org/pla/sites/ala.org.pla/files/content/data/PLA_Services_Survey_Report_2023.pdf
- Goodridge, M., & Rohweder, M. J. (2022). *Librarian's Guide to games and gamers: From collection development to advisory services*. Libraries Unlimited, an imprint of ABC-CLIO, LLC.
- Hill, C. (2016). Play on: The use of games in libraries. *The Christian Librarian*, 59(1). <https://doi.org/10.55221/2572-7478.1157>
- Howarth, J. (2024, January 29). *How many gamers are there? (new 2024 statistics)*. Exploding Topics. <https://explodingtopics.com/blog/number-of-gamers>
- IBM. (n.d.). *IBM® SPSS® Statistics*. IBM.com. <https://www.ibm.com/products/spss-statistics/gradpack>
- Inman Berens, K., & Noorda, R. (2023, November 1). *Gen Z and Millennials: How They Use Public Libraries and Identify Through Media Use*. American Library Association. [https://www.ala.org/advocacy/sites/ala.org.advocacy/files/content/tools/Gen-Z-and-Millennials-Report%20\(1\).pdf](https://www.ala.org/advocacy/sites/ala.org.advocacy/files/content/tools/Gen-Z-and-Millennials-Report%20(1).pdf)
- Lee, V. R., & Phillips, A. L. (2019). *Reconceptualizing libraries: Perspectives from the information and Learning Sciences*. Routledge Taylor & Francis Group.

- Microsoft. (n.d.). *Buy Microsoft Excel (PC or Mac): Cost of Excel only or with Microsoft 365*. Microsoft Store. <https://www.microsoft.com/en-us/microsoft-365/p/excel/cfq7ttc0hr4r?activetab=pivot%3Aoverviewtab>
- Noorda, R., & Inman Berens, K. (2021). Immersive media and books 2020: New insights about book pirates, libraries and Discovery, millennials, and cross-media engagement: Before and during covid. *Publishing Research Quarterly*, 37(2), 227–240. <https://doi.org/10.1007/s12109-021-09810-z>
- Peiser, J. (2024, December 24). *Why board games are quickly regaining popularity* - *The Washington Post*. The Washington Post. <https://www.washingtonpost.com/business/2022/12/24/board-game-popularity/>
- Ponto, K., Tredinnick, R., Verbeke, M., Kopp, K., Swanson, L., & Gagnon, D. (2023). Waddle: Using Virtual Penguin Embodiment as a vehicle for empathy and informal learning. *29th ACM Symposium on Virtual Reality Software and Technology*. <https://doi.org/10.1145/3611659.3617211>
- Post, C., Henry, T., Nunnally, K., & Lanham, C. (2023). A colossal catalog adventure: Representing Indie Video Games and Game Creators in library catalogs. *Cataloging & Classification Quarterly*, 61(7–8), 792–814. <https://doi.org/10.1080/01639374.2023.2280207>
- Robson, D., Phillips, J., & Guerrero, S. (2018). Don't just roll the dice: Simple solutions for circulating tabletop game collections effectively in your library. *Library Resources & Technical Services*, 62(2), 80. <https://doi.org/10.5860/lrts.62n2.80>
- Seay, Jared A. (2024). *Gaming & Libraries*. Jared A Seay - Reference Librarian, Addlestone Library. <https://blogs.cofc.edu/seayj/gaming-learning-articles/library-gaming/>
- Seay, Jared Alexander. (2018a). Considering games and gamification in Libraries & Associated Entities — in praise of real people. *Against the Grain*, 30(5). <https://doi.org/10.7771/2380-176x.8189>
- Seay, Jared Alexander. (2018b). Considering games and gamification in libraries & associated entities — surely, you can't be serious: When library folk go game. *Against the Grain*, 30(2). <https://doi.org/10.7771/2380-176x.8063>
- Seay, Jared Alexander. (2018c). Considering games and gamification in Libraries & Associated Entities — the “g” word. *Against the Grain*, 30(4). <https://doi.org/10.7771/2380-176x.8311>

- Seay, Jared Alexander. (2018d). Considering games and Gamification in libraries & associated entities — how the longing for Tabletops has revitalized games. *Against the Grain*, 30(3). <https://doi.org/10.7771/2380-176x.8268>
- Seay, Jared Alexander. (2021a). Considering games in libraries and such — space guardians: New Year's musings of how libraries of old (including but not necessarily old librarians) have returned just in time to save us from Digital Anxiety. *Against the Grain*, 31(1). <https://doi.org/10.7771/2380-176x.8378>
- Seay, Jared Alexander. (2021b). Considering games in libraries and such — The Ultimate Library instruction. *Against the Grain*, 31(6). <https://doi.org/10.7771/2380-176x.8513>
- Seay, Jared Alexander. (2021c). Considering games in libraries and such — a liminal response to watching paint dry. *Against the Grain*, 31(2). <https://doi.org/10.7771/2380-176x.8386>
- Seay, Jared Alexander. (2021d). Considering games in libraries and such — on using a game to actually teach something. part I: The accidental learner. *Against the Grain*, 31(5). <https://doi.org/10.7771/2380-176x.8501>
- Simplilearn. (2024, March 5). *What is regression analysis? types: Examples: Uses*. Simplilearn.com. <https://www.simplilearn.com/tutorials/excel-tutorial/regression-analysis>

Appendix A: Draft of Games-based Library Services Implementation Questionnaire (Librarians)

Survey Introduction: Thank you for your willingness to participate in this Games-based Library Services Implementation Questionnaire! American libraries' collective assistance with this study will greatly help professionals in the field make informed and forward-thinking decisions related to games-based library services.

At least one staff member from your library that manages programs and/or collections should complete this form; preferably, though, the form would be completed by enough appropriate staff at your library to represent holistic collection and program management that would cater to all library user age demographics across your library space (most often broken down into “children”, “teens”, and “adults”). Participating staff should only fill the form out once.

While completing this questionnaire, it's important for library staff to understand what exactly “games-based library services” [hereto referred to GBLS] means. For the context of our study, “**Games-based**” means any library experiences/offerings directly centered around modern gaming media (including formats like video games, board games, tabletop role-playing games [TTRPGS], virtual and augmented reality games [VR and ARGs], and collectible/competitive card games). “**Library Services**” refers to any library offerings intended to be utilized/experienced by library patrons (like in-person or virtual programs, circulating collections, drop-in/“library use only” collections, informational displays, take-home “passive program” kits, giveaways, and more).

As such, GBLS could be defined as “library experiences/offerings directly centered around modern gaming media like video games, board games, TTRPGs, VR and ARGs, collectible/competitive card games, and other related contemporary gaming formats.” Feel free to use your best judgement about what these services might entail when considering them in the scope of your library's offerings.

Ensuring the ethical integrity of our research is of paramount importance. All responses will be anonymous, and no personally identifiable information will be collected. Data will be stored securely and only accessible to the research team. Any reports or publications resulting from this research will present aggregated data, ensuring individual responses cannot be identified.

If you have any issues/clarifications needed with this questionnaire, or need to alter your responses, please contact the study's Principal Investigator, Chris Baker, at Chris.Baker@dpi.wi.gov. Thank you again for your participation!

Survey Questions:

➤ Library, Community, and Staff Information:

1. Where is your library located?

○ State: _____

- County: _____
 - Municipality: _____
2. What is the population size of your municipality?
 - 0-499
 - 500-4,999
 - 5,000-9,999
 - 10,000-19,999
 - 20,000-49,999
 - 50,000-99,999
 - 100,000+
 3. How would you define your municipality/community? (*Select one*)
 - Rural
 - Urban
 - Sub-urban
 4. What is the size of your library (in square feet; estimated)? _____
 5. How many staff members does your library have (in full-time equivalency, or [FTE](#))? _____
 6. What is your library's total annual programming budget (estimated)?
 7. What is your library's total annual budget for collection materials (estimated)?
 8. What is your role at your library? (*Choose all options that best fit*):
 - Staff serving Youth (*Children's/Youth Services Librarian, Library Assistant, etc.*)
 - Staff serving Teens (*Children's/Youth Services Librarian, Library Assistant, etc.*)
 - Staff serving Adults (*Adult Services Librarian, Library Assistant, etc.*)
 - Staff serving all/multiple demographics (*Programs or Collections Manager, etc.*)
 - Library Director
 - Other: _____
 9. Which of the following library services/functions do you develop, facilitate, and/or manage? (*Choose all that apply*):
 - Library materials collections

- Library programs
- Library displays
- Library budgets
- Community partnership events
- Other: _____

➤ Games-based Library Services Implementations:

1. Does your library currently offer **any** games-based library services (GBLS)?

Note: this could mean games-based programs, collections, displays, or any other service offerings centered around modern game types (listed above).

- Yes
- No

▪ ***If no, skip to question #9***

2. If yes, what formats of GBLS does your library offer, and in what way? (*Multiple choice; select all that apply*):

- Video games
 - Collections for checkout
 - Items for in-library usage
 - In-person programs
 - Virtual programs
 - Displays
- Board games
 - Collections for checkout
 - Items for in-library usage
 - In-person programs
 - Virtual programs
 - Displays
- Tabletop Role-playing Games
 - Collections for checkout
 - Items for in-library usage
 - In-person programs
 - Virtual programs
 - Displays
- Virtual and/or Augmented Reality Technologies
 - Collections for checkout
 - Items for in-library usage

- In-person programs
 - Virtual programs
 - Displays
 - Collectible/competitive card games
 - Collections for checkout
 - Items for in-library usage
 - In-person programs
 - Virtual programs
 - Displays
 - Other (Please describe): _____
3. If your library offers games-based programs, how often are they offered?
- Weekly
 - Monthly
 - Only occasionally throughout the year
4. Approximately how much does your library spend on games-based collections for either checkout or in-library use (*leave blank if none*)? \$ _____
5. Approximately how much does your library spend on either virtual or in-person games-based programs (*leave blank if none*)? \$ _____
6. How does your library fund GBLS? (*Select all the apply*):
- Annual library operating budget
 - Funding from support organization (Friends group, Foundation, etc.)
 - Grants and/or occasional donations
 - Other: _____
7. How does your library promote GBLS to patrons? (*Check all that apply*):
- In-library posters or posters in the community
 - Library displays
 - On our website
 - On social media
 - Through word of mouth at the library
 - We do not promote our GBLS to patrons
8. How popular would you say your GBLS are with your patrons?
- Likert scale: Not popular at all – Extremely popular

9. Why do you not offer GBLS, or what challenges (if any) has your library faced in implementing GBLS or offering more GBLS? (*Choose all options that best fit your experience; leave blank if none*):
- We haven't offered them yet, but now intend to
 - Games are too expensive
 - I feel like I don't know enough about games to integrate GBLS
 - I don't have the time/capacity to integrate GBLS on top of everything else
 - I'm worried that games-based services will not be popular in the future
 - I don't think my community cares about games-based services
 - Libraries shouldn't offer games
 - I think games might be unhealthy for people
 - Other: _____
10. How important do you think it is for the national field of libraries to offer GBLS?
- Likert scale: Not important at all – Very important
11. How personally prepared do you feel to implement, manage, and promote GBLS?
- Likert scale: Not at all prepared – Very prepared
12. If you received professional library education (like a library degree or a library major at a University), or are currently receiving professional library education, have you studied any professional games-based library services in your classes? If yes, please indicate what type(s) (*choose all that apply*):
- Yes
 - Games-based collection development training
 - Games-based collection maintenance training
 - Games-based program development/promotion training
 - Research about games-based library services
 - Information/research about the pervasiveness/popularity of games
 - Other: _____
 - No / Not applicable
 - I don't remember
13. How important do you think it is for modern library training/education programs to teach about GBLS?
- Likert scale: Not important at all – Very important
14. What resources or training would help you feel more prepared to offer GBLS? (*Check all that apply*):
- Virtual webinars, training videos, or webinar/video series
 - In-person conferences or workshops about GBLS

- Digital/analog “toolkits” with ready-to-use resources (posters, social media templates, etc.)
- Guidelines/steps/tips for offering and maintaining games-based programs and/or library collections
- Grants or other funding opportunities to support GBLS
- I don’t need any resources/training related to GBLS
- Other: _____

Appendix B: Draft of Games-based Library Services Needs Assessment Survey (Patrons)

Survey Introduction: Thank you, library user, for your willingness to participate in this “Games-based Library Services Needs Assessment Survey”! In filling out this survey, you’ll be contributing to better national understanding of how library users feel about games-based library services. If you are aware of other people in your community that might be interested in participating in this survey, please share the survey link with them.

Note that participants younger than 18 should get permission from a parent or legal guardian before filling out this survey; additionally, participants aged 11 or younger should only fill this out with direct assistance from a parent or legal guardian. Note that participants should only fill the form out **once**.

While completing this questionnaire, it’s important to understand what exactly we mean when we say “games-based library services” [hereto referred to GBLS]. For the context of our study, “**Games-based**” means any library experiences/offerings directly centered around modern gaming media (including formats like video games, board games, tabletop role-playing games [TTRPGS], virtual and augmented reality games [VR and ARGs], and collectible/competitive card games). “**Library Services**” refers to any library offerings intended to be utilized/experienced by library users (like in-person or virtual programs, circulating collections, drop-in/“library use only” collections, informational displays, take-home “passive program” kits, giveaways, and more).

As such, GBLS could be defined as “library experiences/offerings directly centered around modern gaming media like video games, board games, TTRPGs, VR and ARGs, collectible/competitive card games, and other related contemporary gaming formats.” Feel free to use your best judgement about what these services might entail when considering them in the scope of this survey.

Ensuring the ethical integrity of our research is of paramount importance. All responses will be anonymous, and no personally identifiable information will be collected. Data will be stored securely and only accessible to the research team. Any reports or publications resulting from this research will present aggregated data, ensuring individual responses cannot be identified.

If you have any issues/clarifications needed with this survey, any concerns, or need to alter your responses, please contact the study’s Principal Investigator, Chris Baker, at Chris.Baker@dpi.wi.gov. Thank you again for your participation!

Survey Questions:

➤ Demographics and Library Usage

1. What age range do you belong to?
 - 0-8
 - 8-11

- ☐ 12-18
- ☐ 18-35
- ☐ 35-50
- ☐ 50-65+

2. What is your highest level of education?

- ☐ Pre-K or Kindergarten
- ☐ Grade School
- ☐ Middle School
- ☐ High School/GED
- ☐ College/Trade School
- ☐ Master's Degree/PhD

3. What is your gender identity?

- ☐ Male
- ☐ Female
- ☐ Nonbinary
- ☐ Other: _____
- ☐ Prefer not to answer

4. Which of the following options best describes how frequently you use your library?

- ☐ Daily
- ☐ 2-5 times a week
- ☐ Once a week or so
- ☐ Once or twice a month
- ☐ Occasionally throughout the year, but not frequently

5. What do you use your library for? (*Check all that apply*):

- ☐ Checking out materials
- ☐ Attending library programs
- ☐ As a place to hang out or socialize
- ☐ To get help from library staff with something
- ☐ To use the free wi-fi and/or computers
- ☐ Other: _____

➤ Usage/Interest/Knowledge of Modern Gaming and Libraries:

1. What types of Modern Gaming interest you? (*Check all that apply*):

- ☐ Video games
- ☐ Board games

- ☐ Tabletop Role-playing Games
 - ☐ Virtual Reality
 - ☐ Augmented Reality Games
 - ☐ Collectible/competitive card games
 - ☐ Other: _____
 - ☐ None
2. Does the library you use offer any Games-based library services (GBLS)?; *Note: this could mean games-based programs, collections, displays, or any other service offerings centered around modern game types (listed above).*
- ☐ Yes
 - ☐ No
 - ☐ I don't know
3. Have you ever participated in or used any GBLS at your library?
- ☐ Yes
 - ☐ No
4. If yes, which types of GBLS have you utilized/participated in? (*Check all that apply*):
- ☐ Checked out games-based materials
 - ☐ Attended games-based programs
 - ☐ Participated in gaming competitions
 - ☐ In-library drop-in games offerings (board games, video games, etc.)
 - ☐ Viewed displays and/or educational information posted about gaming
 - ☐ Attended virtual webinars and meetings about games
 - ☐ Other: _____
 - ☐ None
5. What formats of GBLS have you utilized/participated in? (*Check all that apply*):
- ☐ Video games
 - ☐ Board games
 - ☐ Tabletop Role-playing Games
 - ☐ Virtual Reality
 - ☐ Augmented Reality Games
 - ☐ Collectible/competitive card games
 - ☐ Other: _____
 - ☐ None

6. If your library does offer GBLS, do you feel they do an adequate job of promoting them?
- ☐ Yes
 - ☐ No
 - ☐ I don't know
7. Do you wish that your library offered more GBLS?
- ☐ Yes
 - ☐ No
 - ☐ I don't care/I'm not interested in GBLS
8. If yes, what types of GBLS do you wish your library offered more of?
- ☐ Programs centered around gaming
 - ☐ Gaming competitions
 - ☐ Game collections for checkout
 - ☐ In-library usage games
 - ☐ Displays and educational information about gaming
 - ☐ Virtual webinars and meetings about games
9. If your library was to offer more GBLS, what types of games would you like them to focus on/include? *Check all that apply.*
- ☐ Video games
 - ☐ Board games
 - ☐ Tabletop Role-playing Games
 - ☐ Virtual Reality
 - ☐ Augmented Reality Games
 - ☐ Collectible/competitive card games
 - ☐ Other: _____
10. How important do you feel it is for your library to offer GBLS?
- ☐ Likert scale:
 - ❖ Not at all important – Very important
11. How interested are you in participating in future GBLS at your library?
- ☐ Likert scale:
 - ❖ Not at all interested – Very interested